

CENG381 Stochastic Processes -- Mixed Exam

Mixed Exam 5 -- Answer Key

Q1 Solution (30 points)

a) Communicating classes:

Check reachability:

A $\rightarrow$ B (direct), B $\rightarrow$ A (direct): A $\leftrightarrow$ B $\Rightarrow$ {A, B} communicate.

B $\rightarrow$ C (direct), C $\rightarrow$ D (direct), D $\rightarrow$ C (direct): {C, D} communicate.

Can C reach A or B? $P(C,A) = P(C,B) = 0$, $P(D,A) = P(D,B) = 0$: NO.

Can A reach C? A $\rightarrow$ B $\rightarrow$ C: YES (2 steps). But C cannot return to A.

Classes: {A, B} (transient -- can escape to C, D) and {C, D} (recurrent -- closed).

The chain is NOT irreducible (two separate classes).

b) Stationary distribution of recurrent class {C, D}:

Restricted transition matrix:

	C	D
C	[0.7	0.3]
D	[0.4	0.6]

Balance: $\pi(C) = 0.7\pi(C) + 0.4\pi(D) \Rightarrow 0.3\pi(C) = 0.4\pi(D)$

$$\pi(D) = (0.3/0.4)\pi(C) = (3/4)\pi(C)$$

$$\pi(C) + (3/4)\pi(C) = 1 \Rightarrow (7/4)\pi(C) = 1 \Rightarrow \pi(C) = 4/7$$

$$\pi(D) = 3/7$$

Check: $4/7 + 3/7 = 1$ (OK)

c) Number of steps for convergence:

Solve $(1/2)*(0.3)^n < 0.01$:

$$(0.3)^n < 0.02$$

$$n * \log(0.3) < \log(0.02)$$

$$n > \log(0.02) / \log(0.3) \quad [\text{inequality flips because } \log(0.3) < 0]$$

$$n > (-3.912) / (-1.204) \sim 3.25$$

So $n \geq 4$ steps is sufficient. After 4 steps the total variation distance to stationarity is below 0.01.

Q2 Solution (35 points)

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a) Leaving rates and embedded chain P:

$$q(I) = 3.0, \quad q(P) = 4.0, \quad q(C) = 2.0$$

Embedded chain (divide off-diagonal entries by leaving rate):

	I	P	C	
I	[0	1.0	0	] (only jump: I->P)
P	[1/4	0	3/4	]
C	[1.0	0	0	] (only jump: C->I)

b) Stationary distribution of the CTMC:

Global balance equations (rate out = rate in):

$$\begin{aligned} 3\pi(I) &= 1\pi(P) + 2\pi(C) && \dots \text{ (I)} \\ 4\pi(P) &= 3\pi(I) && \dots \text{ (II)} \\ 2\pi(C) &= 3\pi(P) && \dots \text{ (III)} \end{aligned}$$

From (II): $\pi(P) = (3/4)\pi(I)$

From (III): $\pi(C) = (3/2)\pi(P) = (3/2)(3/4)\pi(I) = (9/8)\pi(I)$

Normalise: $\pi(I) + (3/4)\pi(I) + (9/8)\pi(I) = 1$

$$\begin{aligned} \pi(I) * (8/8 + 6/8 + 9/8) &= 1 \Rightarrow \pi(I) * (23/8) = 1 \\ \pi(I) &= 8/23, \quad \pi(P) = 6/23, \quad \pi(C) = 9/23 \end{aligned}$$

Check: $8 + 6 + 9 = 23$ (OK)

c) Long-run availability:

$$\begin{aligned} \text{Fraction of time available} &= \pi(I) + \pi(P) = 8/23 + 6/23 = 14/23 \\ &\sim 0.609 \end{aligned}$$

The server is available (Idle or Processing) about 60.9% of the time.

Q3 Solution (35 points)

a) Martingale classification:

- (i) $S(n)$ is a martingale iff $E[X(i)] = 0$, i.e., $p = q = 1/2$.
- (ii) $S(n)^2 - n$ is a martingale iff $p = q = 1/2$ and $\text{Var}[X(i)] = 1$ (which holds since steps are ± 1).
- (iii) $(q/p)^{S(n)}$ is a martingale for ANY $p \neq 1/2$ (De Moivre's martingale). When $p = 1/2$, $r = q/p = 1$ and the martingale is trivially 1 everywhere.

b) Win probability and expected game length:

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OST on $S(n)$ (symmetric, T bounded):

$$\begin{aligned}E[S(T)] &= S(0) = 3 \\7 * h(3) + 0 * (1 - h(3)) &= 3 \\h(3) &= 3/7 \sim 0.429\end{aligned}$$

OST on $S(n)^2 - n$:

$$\begin{aligned}E[S(T)^2 - T] &= S(0)^2 - 0 = 9 \\E[S(T)^2] &= 49 * (3/7) + 0 * (4/7) = 21 \\21 - E[T] &= 9 \Rightarrow E[T] = 12 \text{ steps}\end{aligned}$$

c) Modified stopping time $T' = \min(T, T_5)$:

T' is a stopping time (minimum of two stopping times is a stopping time).

T' is bounded: $T' \leq T \leq N^2 = 49$ steps (it can be shown T is a.s. finite).

The increments $|S(n+1) - S(n)| = 1$ are bounded.

All OST conditions are satisfied, so $E[S(T')] = S(0) = 3$.

At stopping time T' :

$S(T') = 5$ if the player reaches 5 before 0 or 7 (prob = $P(T_5 < T)$)

$S(T') = 7$ if the player reaches 7 (bypassing 5 -- not possible going from 3 to 7; they MUST pass through 5 first since steps are ± 1)

Actually: to go from 3 to 7, the player MUST pass through 5.

So $T' = T_5$ whenever $h(3) > 0$, meaning the player always hits 5 on the way to 7 (but may also hit 5 and then eventually reach 0).

More precisely: $E[S(T')] = 5 * P(\text{reach 5 before 0}) + 0 * P(\text{reach 0 before 5}) = 3$

$\Rightarrow P(\text{reach 5 before 0} \mid \text{start 3}) = 3/5$.

Interpretation: the OST value $E[S(T')] = 3$ encodes the probability of reaching the nearer boundary (5 or 0) directly -- showing that the probability of reaching 5 before 0 is $3/5$, as expected by symmetry for a symmetric walk with barrier at 5 and ruin at 0 starting from 3.